

A Field Demonstration of Cross Lake Free Space Optical Communication for 10GPON Backhaul

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Abstract This paper demonstrates a field trial in PON network utilizing a custom-designed FSO system cross an 1.3km lake environment. Operating at C-band without OA and FEC, the system achieves consecutive 6 hours of error-free 10Gbps transmission, validating FSO as an alternative solution for operator backhaul. ©2026 The Author(s)

Introduction

Free space optical communication (FSOC) has seen a resurgence of interest with the rapid evolution of network infrastructure, particularly for inter-satellite links and emerging terrestrial applications [1]. However, the deployment of FSOC in terrestrial environments faces unique challenges compared to space-based systems. Severe atmospheric channel conditions have historically limited its widespread adoption as a primary solution for near-ground networks [2,3]. This has led to FSOC being largely considered a niche technology for terrestrial applications, despite its inherent advantages of high bandwidth, license-free operation, and low latency.

The operators and service providers continue to seek reliable and cost-effective alternatives to fiber for last-mile connectivity and backhaul, especially in difficult terrains like rivers, mountains, or lakes where fiber deployment is expensive or impractical. However, a robust, commercially viable, and easily deployable FSOC system capable of sustaining high data rates in such challenging near-ground conditions for existing operator networks remains a critical unmet need [4]. Prior demonstrations often involve complex setups or do not specifically target integration with operators' real network such mobile fronthaul/backhaul or PON [5,6].

Recommendation ITU-T G.641 has been published to specify the short range terrestrial application physical layer interoperability [7]. But no commercial products have been reported to demonstrate a practical deployment.

In this paper, we designed a commercially ready FSOC system for PON backhaul network, analyzed the system capture probability and link margin, provided the field trial test results, and finally drew the conclusion and prospective of FSOC.

System configuration and key designs

The custom-designed system configuration is

shown in Fig.1. It consists of four sub components: an optical-mechanical system, a fine tracking system, a coarse tracking system (reserved), and a communication system. The telescope is responsible for beam expansion of the light spot, while the coarse tracking detector captures the signal. The fine tracking detector and fine tracking actuator perform real-time beam tracking, emission, and reception coupling. The transmission/reception unit handles collimated beam emission and signal coupling. It should be noted that we reserve the optical amplifier (OA) function in the light path for long distance applications but it is not necessary in this scenario. The whole system is data and protocol transparent, which strictly follows the specification of G.641 and simplifies the data processing.

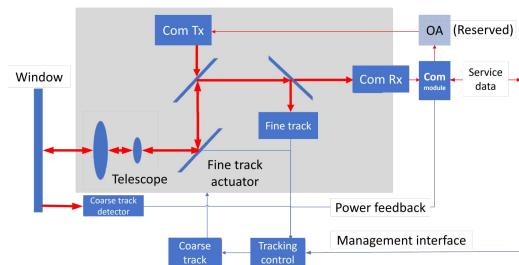


Fig.1: System configuration. Com Tx and Rx are communication system's transmitting and receiving part.

For the installation of an FSOC system across lakes, the design specifications require that the field of view (FOV) angle of the coarse tracking detector, the telescope's FOV angle, and the optical stroke of the fine tracking actuator larger than the maximum vibration angle to make sure the light captured inside the telescope. After coarse tracking, the target enters the fine tracking field of view, the fine tracking detector detects the light spot, initiates closed-loop fine tracking control, stabilizes the tracking trajectory, and notifies successful chain establishment to commence communication. The tracking control board is designed with reserved high-

performance computing units to run the network management system, enabling users to monitor and control the operational status of the communication system through this platform.

The telescope employs an internally unfocused Galilean-type beam expander. Externally, it compresses the transmitted parallel light and introduces it into the rear optical path; internally, it expands the collimated light from the rear optical path, compresses the divergence angle, and then emits the light. The coarse tracking camera lens employs a three-element lens design. The effective focal length is approximately 56 mm, with the system's Abbe patch radius within $21.2 \mu\text{m}$, corresponding to a detector imaging size of 4 pix*4 pix to 5 pix*5 pix. Additionally, a custom narrow band is added to the front of the tracking lens to suppress stray light, thereby enhancing the isolation of signal light emission. The precision tracking camera lens also employs a three-element lens structure. The effective focal length is approximately 25 mm. The lens has an effective aperture of 12mm, also with a custom narrow band added. For the communication subsystem, multimode fiber is used as the receiving fiber, while single-mode fiber serves as the transmitting fiber. The lens element of the transmission module employs two spherical surfaces, achieving 4 mm focal length while for the receiving part, a single-piece aspherical lens is used to achieve 23.2 focal length. The four subsystem optical design is conceptually illustrated in Fig.2.

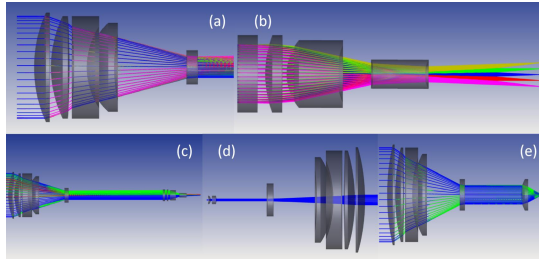


Fig.2: Subsystem design diagrammatic figure: (a) Telescope; (b) Coarse tracking; (c) Fine tracking; (d) Tx lens; (e) Rx lens.

After all the system analysis, we design a 60mm diameter telescope to achieve $\pm 2^\circ$ coarse tracking FOV and 4 mrad fine tracking FOV. The divergence of emission light is set at $480 \mu\text{rad}$ while the reception FOV is $530 \mu\text{rad}$. The bidirectional operating wavelengths are chosen as $1563.05 \pm 0.2 \text{ nm}$ and $1540.56 \pm 0.2 \text{ nm}$, leaving 23 nm gap to avoid channel crosstalk. By this design, we achieved a full capture range in targeted PON environment with the ability of covering 5km distance with maximum data speed of 10Gbit/s, even without OA and FEC. This design is a commercially ready solution up to our acknowledgement, in operator's real

network, replicable and promotable.

System capture probability and link margin analysis

The capture probability can be expressed as $P_{cap} = P_U P_S P_D$, where P_U is the coverage rate of the target FOV for the uncertain region, P_S is the FOV coverage probability during scanning and P_D is the the detection probability. For the designed parameters aforementioned, P_U and P_S reaches 100%. P_D is related to the detection current threshold set by the photon detector. The beacon light incident on the detector, background light, and detector noise all generate currents. If the detector output current exceeds the threshold current, it indicates that a valid signal has been received, and the opposing beacon has been captured. If the current generated by background noise and interference exceeds the threshold current, false alarms will occur. Assuming mean noise value $T_n \delta_n$ as the threshold and the signal intensity as $T_n \delta_n + T_t \delta_t$, in which δ_n is standard deviation of noise signal and δ_t is the standard deviation of signal when available. For a single pixel, the expressions for false alarm probability and loss probability are as follows:

$$P_{fa} = \frac{1}{2} \text{erfc}\left(\frac{T_n}{\sqrt{2}}\right)$$

$$P_{lp} = \frac{1}{2} \text{erfc}\left(\frac{T_t}{\sqrt{2}}\right)$$

If the light spot covers $M \times M$ pixels, the frame-level missed detection probability is $P_{LP} = P_{lp}^{M \times M}$. If the number of pixels on the detector not covered by light spots is $N \times N$, the scanning capture process spans n frame periods, with a false alarm probability at the frame level of $P_{FA} = 1 - (1 - P_{fa})^{nN \times N}$. In this demonstration, a moderate value set of $T_n = 7$, $T_t = 8$, $M=9$, $N=400$ and $n=5000$ is chosen, thus the P_D at the frame level is $P_D = (1 - P_{LP})(1 - P_{FA}) = 99.65\%$. Therefore, the P_{cap} is 99.65%, almost 20% higher than the reported ones [2].

Next, a link margin analysis is performed. We assume a 5km span with transmitting power of 0.5W, 70% emission efficiency. The visibility sets at 5km as well. The reception efficiency (light coupled from free space to lens than to the fibre before finally into the detector) is 45% and the receiver sensitivity (commercial 10G DWDM transceiver) is -26dBm. For this distance, if OA is used with 26.99 dB gain, considering pointing error and necessary free space link loss, the link margin is 23.24 dB, and if OA is not used, the margin remains 2.76dB.

Field test results

In this demonstration, the Broadband Remote Access Server (BRAS) end transceiver operates at 1550 nm band while the Optical Line Terminal (OLT) at 1310 nm band. So a conversion from client to line side is necessary. We use commercial 10G 40KM 1550nm transceiver for BRAS end and 10G 10KM 1310nm transceiver for OLT end. The BRAS site is located 40 km away from the lake shore FSO device (FSO 1) while the OLT is located on the island about 100 meters away from the FSO device (FSO 2), seen as in Fig.3 (a). A pair of commercial DWDM 10G transceivers are used for line side. The real environment is shown in Fig.3 (b).

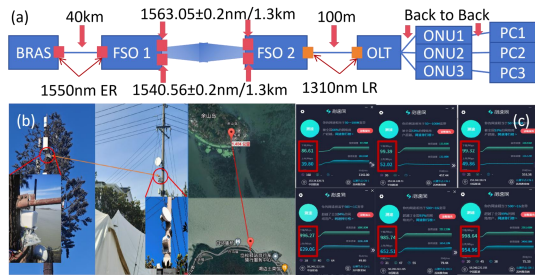


Fig.3: (a) the practical setup of this demonstration. (b) the real deployment environment for both sides, the zoomed in figures show the device appearance. (c) the speed test results for 3 PCs using radio (upper bracket) and FSO (lower bracket) backhauling.

Since the system is data and protocol transparent, so we monitor the received power and BER at line side. And we evaluate the PON service with 3 computers linking to individual optical network units (ONUs). The 3 PCs have identical network cards whose maximum speed is 1000Mbps for each. The received power and accumulated BER over time on island and shore are depicted in Fig.4 (a) & (b). In the morning, the temperature is steady so the vapour from the lake changes slowly, the received power keeps at a relatively small changing range. With the heat accumulation, the vapour from the lake reaches summit, fluctuations of refractive index result in fluctuations of receiving power, thus BER has a sudden rise, although it is still under threshold. In the afternoon, the fluctuations intensifies but the accumulated BER rises slowly because the fast changing steering mirror keeps the Tx and Rx centers aligned on the same axis. The corresponding island side device's fine tracking system's X and Y axis locations are recorded along with the BER and are depicted in Fig.4 (c) & (d). From the location variations, one can tell that under frequent movement of tower, the tracking system responses real time to achieve the light confined within the tracking window, realizing the designed ability to prevent micro vibration. The 3 PCs achieved 998Mbps/965Mbps/996Mbps respectively

bearing real PON service. Based on the same demonstration condition, data rates using radio backhauling are only 86Mbps/99Mbps/99Mbps, respectively at the meantime. Hence, the data rate increases more than 10 times compared to radio backhaul.

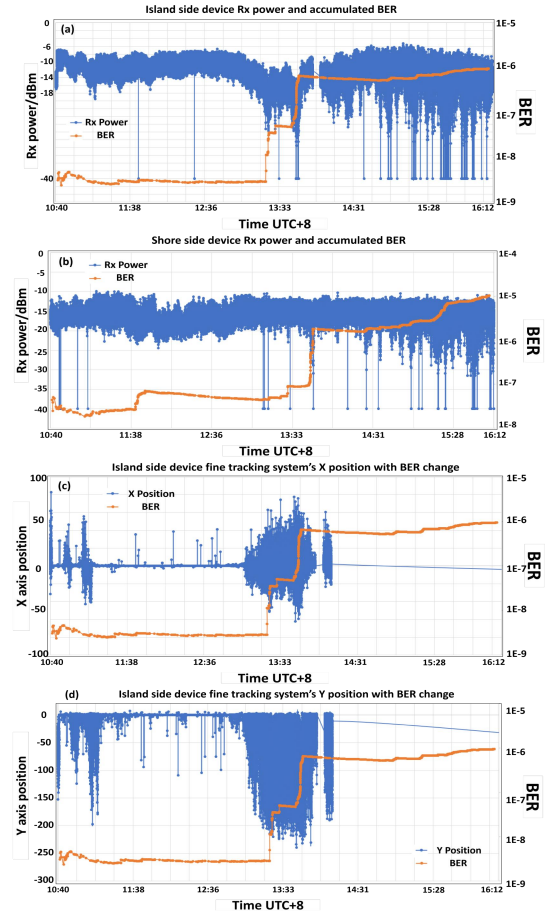


Fig.4: (a)&(b) Island and shore side device line side Rx power and accumulated BER over 6 hours time. (c)&(d) Island side device fine tracking system's X and Y axis position change with BER over 6 hours times.

Conclusions

This paper demonstrates a field trial of a custom-designed FSO system for 10GPON backhaul cross an 1.3km lake environment. Utilizing an amplifier-free C-band short range solution, the system achieved robust, error-free 10Gbps transmission for consecutive 6 hours, effectively mitigating environmental fluctuations and delivering a significant ~10-fold increase in network speed for island users compared to microwave backhaul. This demonstration significantly advances the case for FSO as a high-capacity, reliable, and commercially viable alternative to traditional fiber or microwave links in difficult terrestrial terrains, paving the way for its broader adoption in operator networks.

Acknowledgements

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